

AAYUSHA CHAKRABORTY

Data Scientist | ML · Forecasting · Supply Chain Analytics

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SUMMARY

Data Scientist with 4+ years building production ML systems for supply chain, demand forecasting, and pricing across manufacturing and e-commerce. Strong in Python, SQL, time-series, XGBoost/LightGBM, reinforcement learning, and MLOps (MLflow, Docker, AWS). Top-5 finalist in 24+ national data science competitions. Translates ambiguous business problems into measurable revenue, margin, and efficiency gains.

TECHNICAL SKILLS

Languages & Databases: Python (Pandas, NumPy, Scikit-learn, Statsmodels), SQL (PostgreSQL, MySQL)

Machine Learning: XGBoost, LightGBM, Random Forest, Logistic/Linear Regression, K-Means, SHAP, Feature Engineering

Forecasting & Time Series: ARIMA, SARIMA, Prophet, Exponential Smoothing, Holt-Winters, Hierarchical Forecasting

Deep Learning & RL: TensorFlow/Keras, Reinforcement Learning (RLlib, PPO), LSTM, Multi-Agent Systems (Mesa)

MLOps & Cloud: MLflow, Docker, Airflow, Kedro, AWS (EC2, S3, Lambda), FastAPI

Visualization & BI: Power BI, Tableau, Excel, Plotly, Matplotlib, Seaborn, React + Recharts

PROFESSIONAL EXPERIENCE

Data Scientist | Jayanita Exports (Manufacturing & Exports)

Jan 2025 – Present

- Built **SKU-level demand forecasting pipeline** (Prophet + LightGBM) on 18 months of order data; **forecast MAPE 24% → 11%**, inventory holding cost down 18%.
- Developed **supplier risk scoring model** (XGBoost on lead-time, defect rate, OTD); flagged top-15% risk suppliers — 8% raw-material cost reduction.
- Designed **anomaly detection system** for QC metrics (Isolation Forest + SPC); cut defect-driven return rates by 15%.
- Built CM3 / SKU profitability models — 12% margin improvement across top-50 products. Automated KPI reporting (Python + Power BI), **reducing weekly effort by ~60%**.

Data Analyst | Tastes2Plate (E-commerce / D2C Food)

Jan 2024 – Jan 2025

- Built time-series + regression forecasting models (SARIMA, Prophet, XGBoost) across 300+ SKUs; **stockouts down 35%, overstock down 22%**.
- Developed dynamic pricing & discount optimization model (price-elasticity regression); **18% GMV lift** on 40-SKU pilot.
- RFM segmentation + churn modeling; reactivation campaigns lifted 90-day repeat-purchase rate by 22%. Power BI dashboards adopted for daily founder standups.

Marketing Analyst | Rankkking (Digital / Web3)

Jan 2021 – Dec 2022

- Built Marketing Mix Models + multi-touch attribution; **ad-spend efficiency +25%, ROI +20%**. A/B testing framework for 12+ campaigns; +82% organic traffic, +15% conversions; supported clients scaling to 300K+ MAU.

KEY PROJECTS

DataNiraya — Supply Chain Intelligence Platform · React, Vite, Tailwind, Recharts, FastAPI, Python, XGBoost

- Building a **full-stack SaaS platform** unifying demand forecasting, inventory optimization, supplier risk, and anomaly detection in one executive workspace. React + Vite + Tailwind + Recharts frontend with modular API abstraction layer; FastAPI ML microservices for Prophet/LightGBM forecasting, Isolation Forest anomaly, and XGBoost supplier scoring.

Dynamic Pricing Engine — RL · Python, RLlib (PPO), Prophet, Scikit-learn

- RL pricing agent (PPO) adjusting prices on demand elasticity, competitor moves, and inventory; **12–18% revenue uplift** over rule-based baseline in 12-month backtests. Custom Gym env with Prophet forecasts as state features.

Supply Chain Disruption Prediction · Python, XGBoost, LightGBM, Statsmodels

- ML pipeline predicting supplier delays from lead-time, weather, port-congestion, and reliability features. **0.87 ROC-AUC**; SHAP-driven feature analysis reduced simulated disruption costs 15–20%.

Multi-Agent Business Strategy Simulator · Python, Mesa, RLlib, Kedro, MLflow

- Digital-twin simulation with 5 RL agents (Finance, Procurement, Sales, HR, Operations); compressed strategic planning cycles from weeks to minutes. Kedro + MLflow for reproducible experiment tracking.

ACHIEVEMENTS, EDUCATION & CERTIFICATIONS

- Top-5 finalist in 24+ national data science competitions. DecodeX 2024 — Top 4:** end-to-end ML pipeline (650-variable feature selection, XGBoost, SHAP) for loan approval; +9 pts accuracy. **Renaissance 2024 — 3rd Place:** predictive + optimization framework for municipal budget allocation.

MBA, Business Analytics — Sharda University · **BBA, Marketing** — Kalinga University

CGPA: 8.4 / 9.14

Certifications: Data Science & ML (NSDC, BCG-Forge) · AWS Fundamentals (Coursera) · Power BI (PwC) · Data Visualization (Accenture, TATA)